
Linear Programming

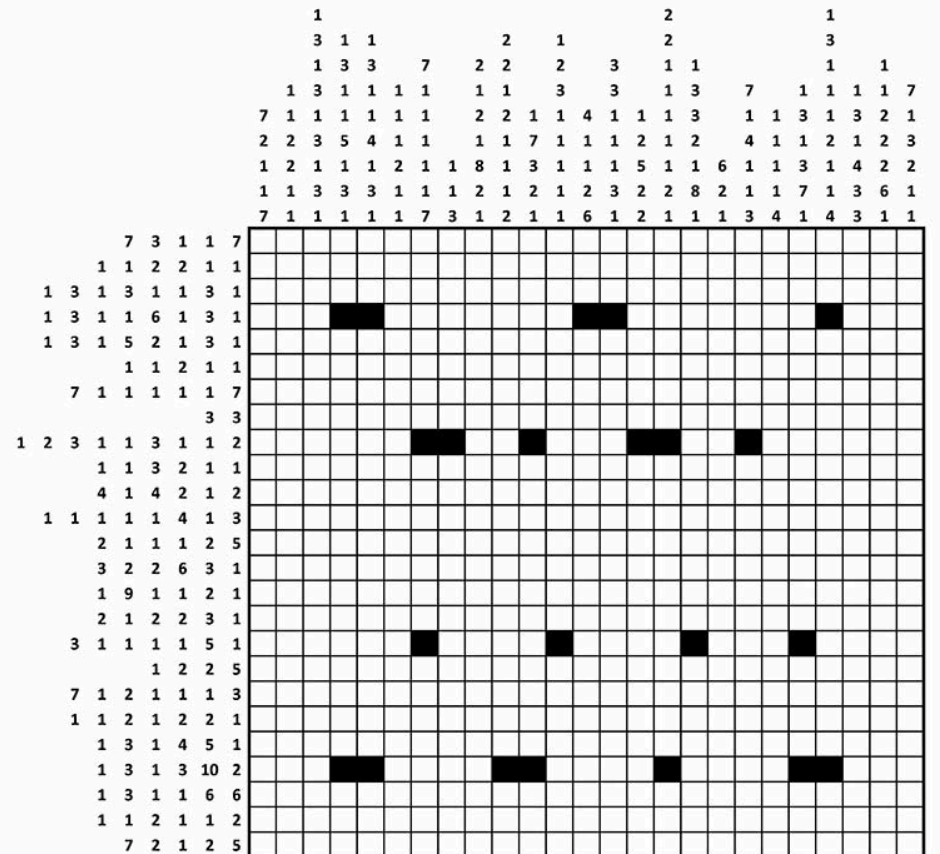
CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



TODAY

- Solving Linear Program
- Integer Program
- Branch and Bound



GCHQ



LINEAR PROGRAMMING: WHAT TO EAT?

We are trying healthy by finding the optimal amount of food to purchase.

We can choose the amount of **stir-fry** (ounce) and **boba** (fluid ounces).

Health Goals

- $2000 \leq \text{Calories} \leq 2500$
- $\text{Sugar} \leq 100g$
- $\text{Calcium} \geq 700mg$

Food	Cost	Calories	Sugar	Calcium
stir-fry (per oz)	1	100	3	20
Boba (per fl oz)	0.5	50	4	70

What is the cheapest way to stay "healthy" with this menu?

How much **stir-fry** (ounce) and **boba** (fluid ounces) should we buy?

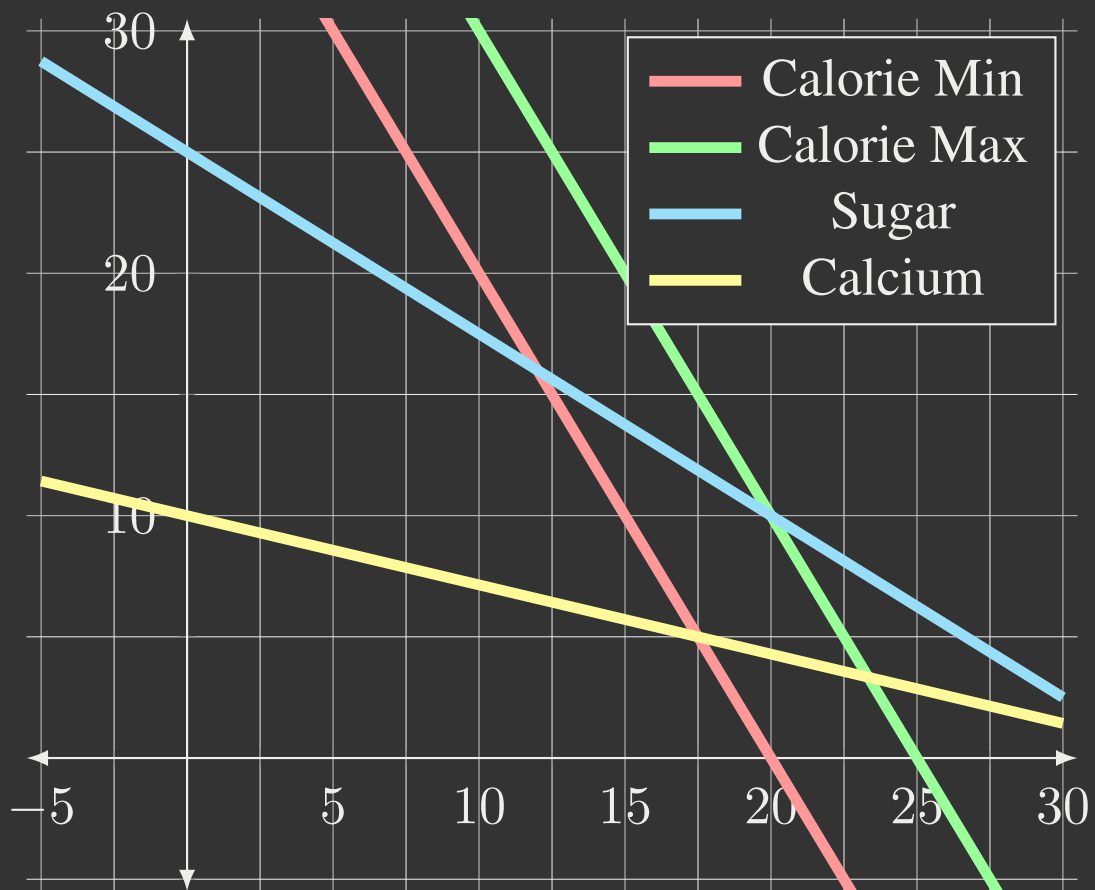


OPTIMIZATION FORMULATION

Diet Problem

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{A}\mathbf{x} \leq \mathbf{b}\end{array}$$

$$A = \begin{bmatrix} -100 & -50 \\ 100 & 50 \\ 3 & 4 \\ -20 & -70 \end{bmatrix} \quad b = \begin{bmatrix} -2000 \\ 2500 \\ 100 \\ -700 \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$



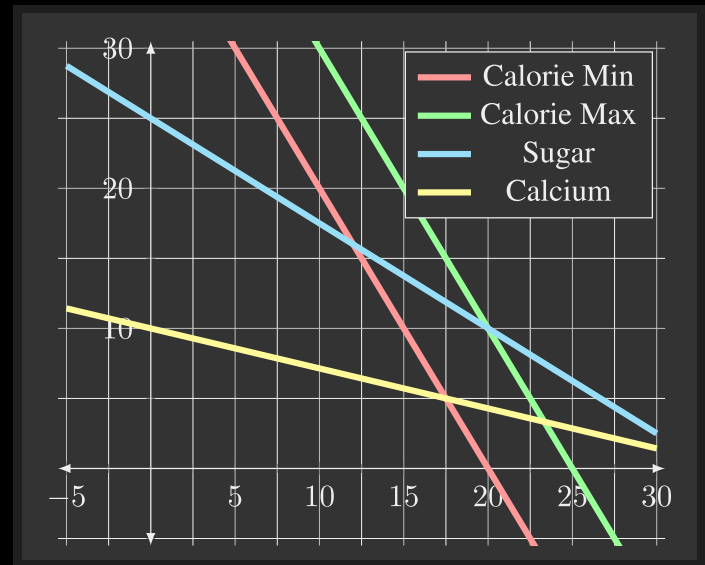
REPRESENTATION AND PROBLEM SOLVING

Problem Description

Optimization Representation

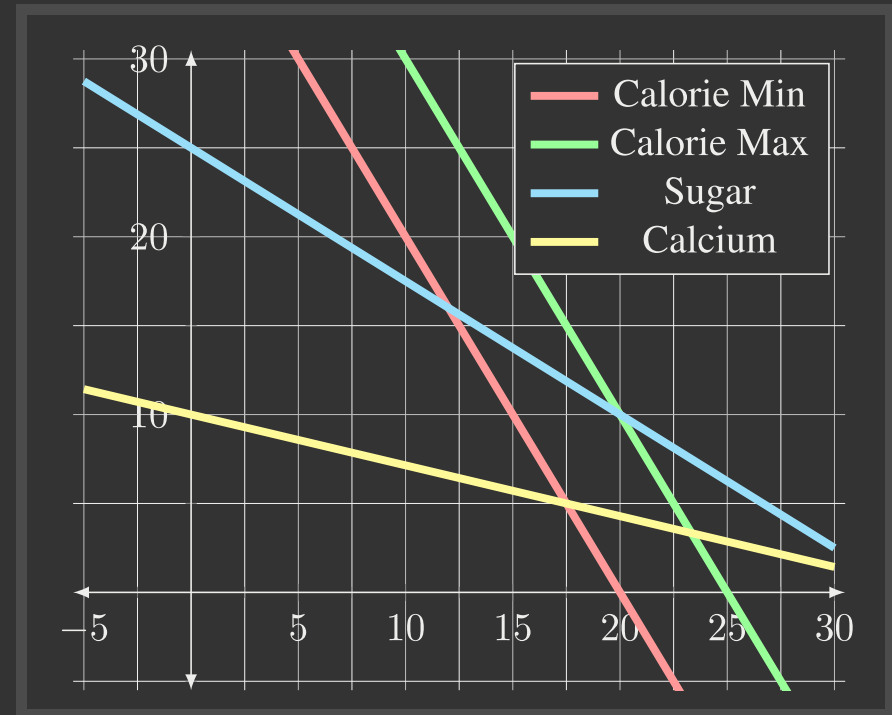
$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

Graphical Representation



SOLVING A LP

Solutions are at feasible intersections of constraint boundaries.



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Algorithms

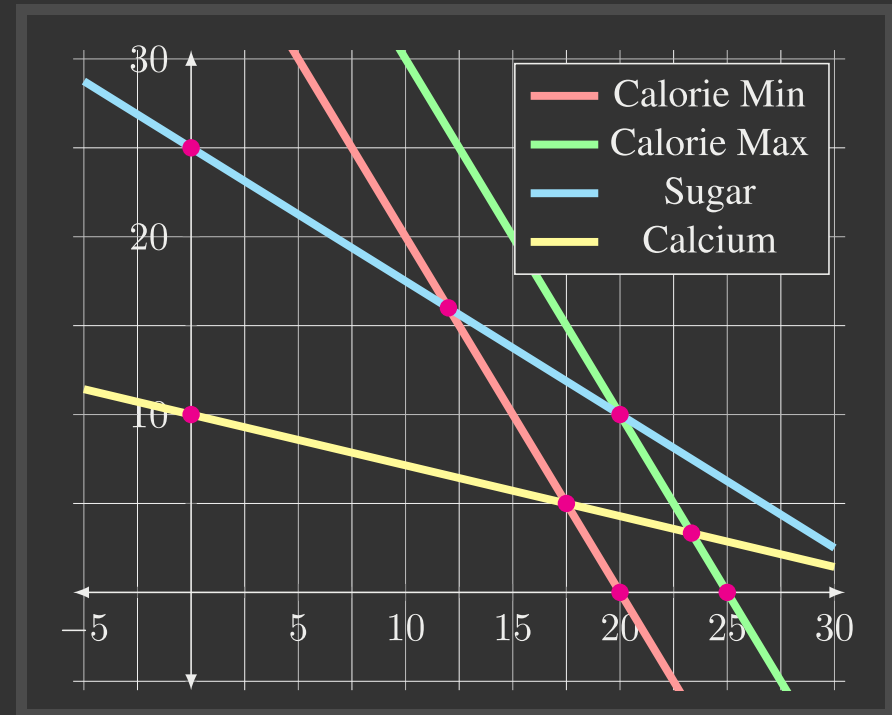


SOLVING A LP

Solutions are at feasible intersections of constraint boundaries.

Algorithms

- Check objective at all feasible intersections.

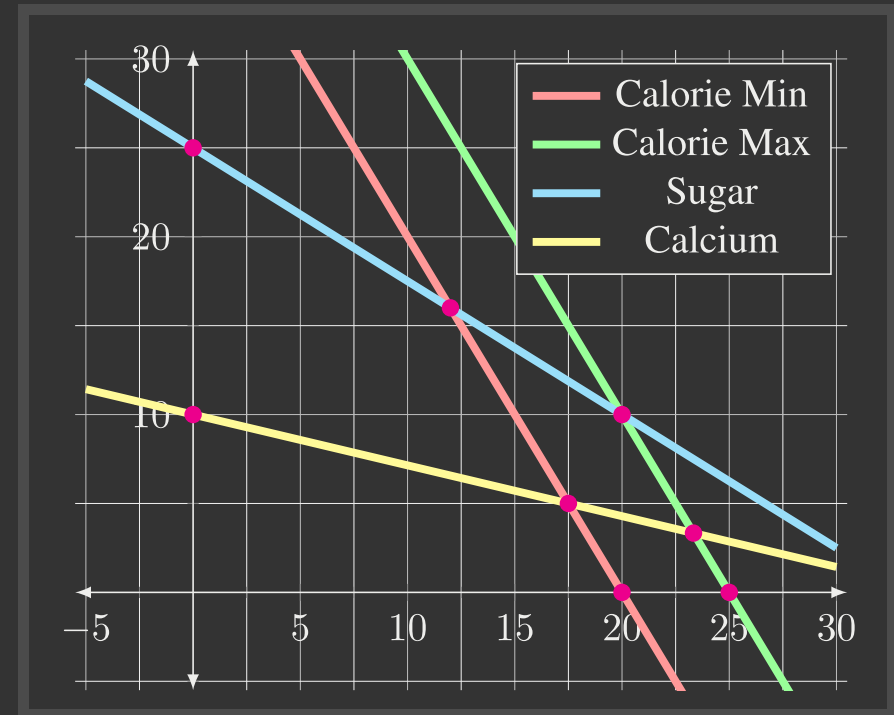


SOLVING A LP

Solutions are at feasible intersections of constraint boundaries.

Algorithms

- Check objective at all feasible intersections.
- Simplex

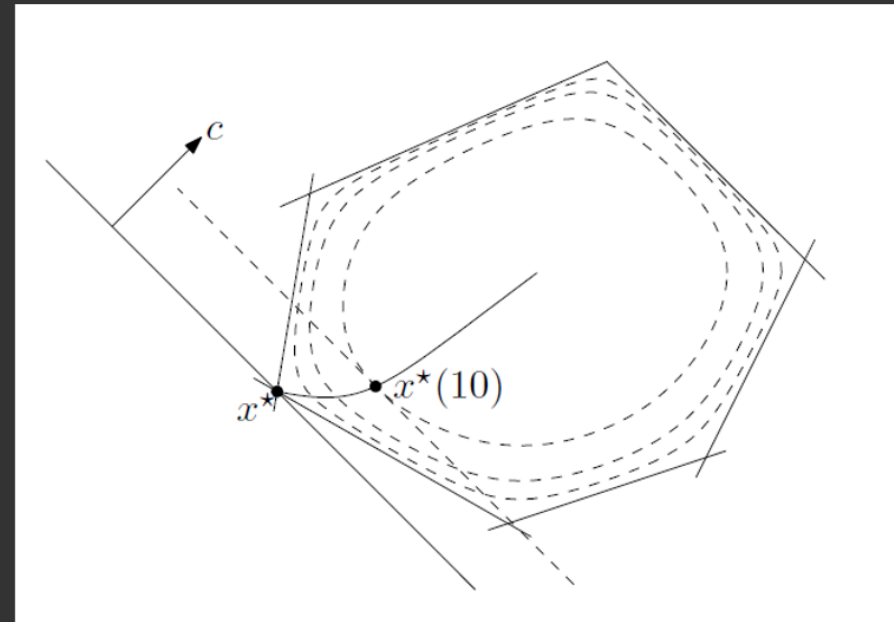


SOLVING A LP

Solutions are at feasible intersections of constraint boundaries.

Algorithms

- Check objective at all feasible intersections.
- Simplex
- Interior Point



SOLVING AN LP

But, how do we find the intersection between boundaries?

$$\begin{array}{ll} \min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \leq \mathbf{b} \end{array} \quad A = \begin{bmatrix} -100 & -50 \\ 100 & 50 \\ 3 & 4 \\ -20 & -70 \end{bmatrix} \quad b = \begin{bmatrix} -2000 \\ 2500 \\ 100 \\ -700 \end{bmatrix} \quad \text{and } c = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$



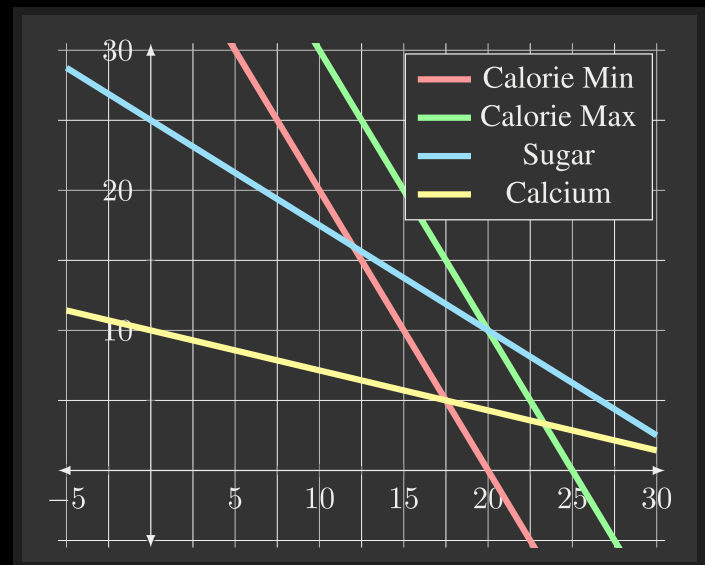
WHAT ABOUT HIGHER DIMENSIONS?

Problem Description

Optimization Representation

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

Graphical Representation



SHAPES IN HIGHER DIMENSIONS

How do these linear shapes extend to 3-D, N-D?

2-D

3-D

N-D

$$a_1x_1 + a_2x_2 = b_1$$

$$a_1x_1 + a_2x_2 \leq b_1$$

$$a_{1,1}x_1 + a_{1,2}x_2 \leq b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 \leq b_2$$

$$a_{3,1}x_1 + a_{3,2}x_2 \leq b_3$$

$$a_{4,1}x_1 + a_{4,2}x_2 \leq b_4$$



SHAPES IN HIGHER DIMENSIONS

How do these linear shapes extend to 3-D, N-D?

	2-D	3-D	N-D
$a_1x_1 + a_2x_2 = b_1$	line	plane	hyperplane
$a_1x_1 + a_2x_2 \leq b_1$	halfplane	halfspace	halfspace
$a_{1,1}x_1 + a_{1,2}x_2 \leq b_1$	polygon	polyhedron	polytope
$a_{2,1}x_1 + a_{2,2}x_2 \leq b_2$			
$a_{3,1}x_1 + a_{3,2}x_2 \leq b_3$			
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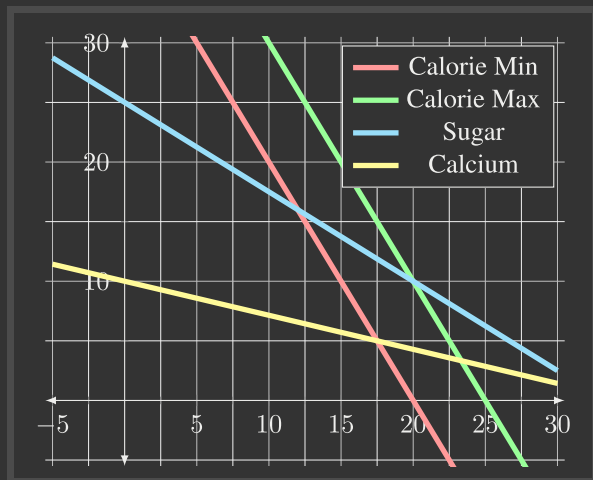
INTERSECTIONS IN HIGHER DIMENSIONS

How do these linear shapes extend to 3-D, N-D?

$$\begin{array}{ll}\min & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

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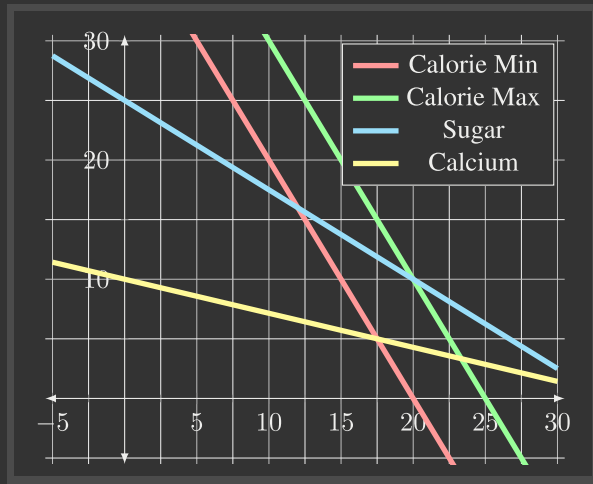


INTERSECTIONS IN HIGHER DIMENSIONS

Still looking at subsets of $\mathbf{A}$ matrix

$$\begin{array}{ll} \min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{array}$$

$$\mathbf{A} = \begin{bmatrix} -100 & -50 \\ 100 & 50 \\ 3 & 4 \\ -20 & -70 \end{bmatrix}$$



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LINEAR $\rightarrow$ INTEGER

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LINEAR VS INTEGER PROGRAMMING

Linear objective with linear constraints, but now with additional constraint that all values in $\mathbf{x}$ must be integers

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^N\end{array}$$



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We could also do:

- Even more constrained: Binary Integer Programming
- A hybrid: Mixed Integer Linear Programming



INTEGER PROGRAMMING: GRAPHICAL REPRESENTATION

Just add a grid of integer points onto our LP representation

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^N\end{array}$$



RELAXATION

Relax IP to LP by dropping integer constraints

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^N\end{array}$$

Remember Heuristics?



SOLUTION OF LINEAR VS INTEGER PROGRAM

Let y_{IP}^* be the optimal objective of an integer program P .

Let $\mathbf{x}_{IP}^*$ be the optimal point of an integer program P .

Let y_{LP}^* be the optimal objective of the LP-relaxed version of P .

Let $\mathbf{x}_{LP}^*$ be the optimal point of the LP-relaxed version of P .

Assume that P is a minimization problem.

Which of the following are true?

- $\mathbf{x}_{IP}^* = \mathbf{x}_{LP}^*$
- $y_{IP}^* \leq y_{LP}^*$
- $y_{IP}^* \geq y_{LP}^*$

$$\begin{aligned} y_{IP}^* = \min_{\mathbf{x}} \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^N \end{aligned}$$

$$\begin{aligned} y_{LP}^* = \min_{\mathbf{x}} \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \end{aligned}$$

SOLVING INTEGER PROGRAMS: A NAIVE SOLUTION

True/False: It is sufficient to consider the integer points around the corresponding LP solution?



SOLVING AN IP

Branch and Bound algorithm



SOLVING AN IP

Branch and Bound algorithm

- Start with LP-relaxed version of IP



SOLVING AN IP

Branch and Bound algorithm

- Start with LP-relaxed version of IP
- If solution $\mathbf{x}_{LP}^*$ has non-integer value at $\mathbf{x}_i$
 - Consider two branches with two different slightly more constrained LP problems:
 - Left branch: Add constraint $\mathbf{x}_i \leq \text{floor}(\mathbf{x}_{LP}^*)$
 - Right branch: Add constraint $\mathbf{x}_i \geq \text{ceil}(\mathbf{x}_{LP}^*)$



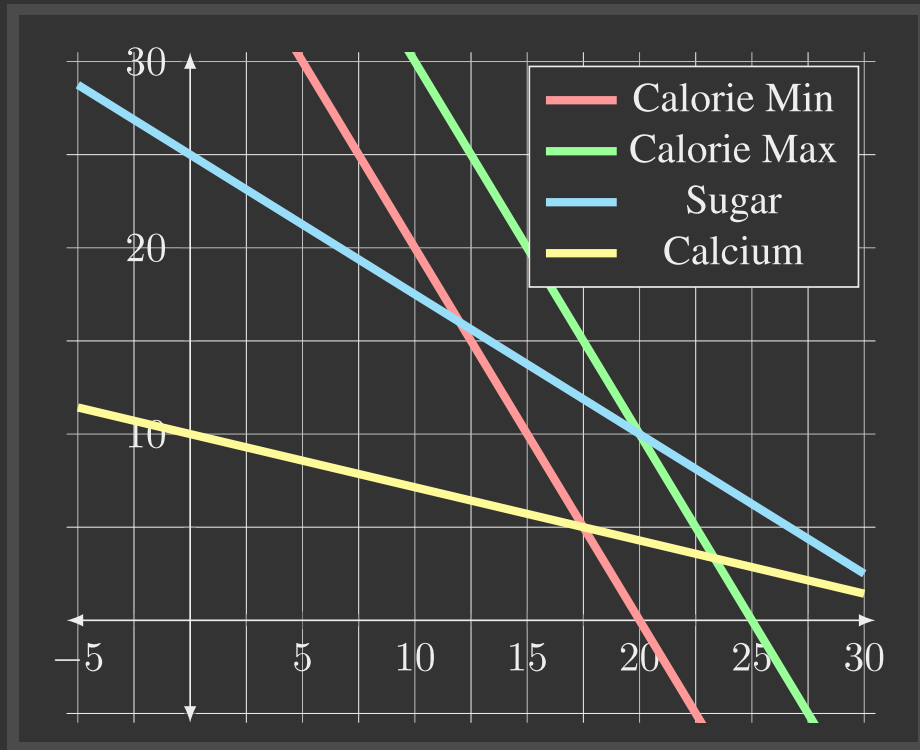
SOLVING AN IP

Branch and Bound algorithm

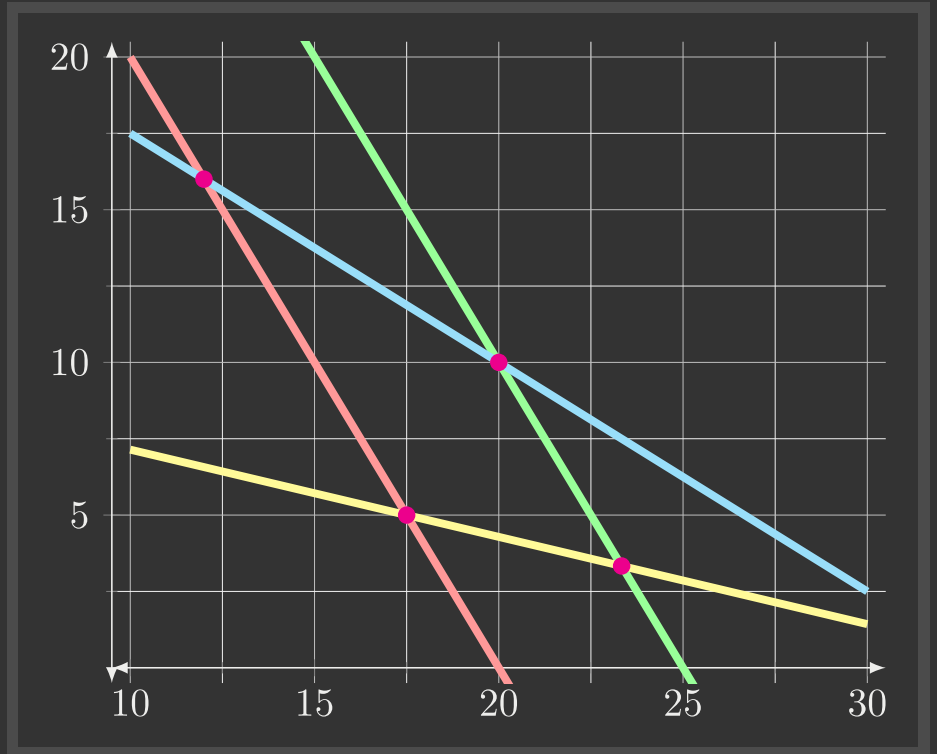
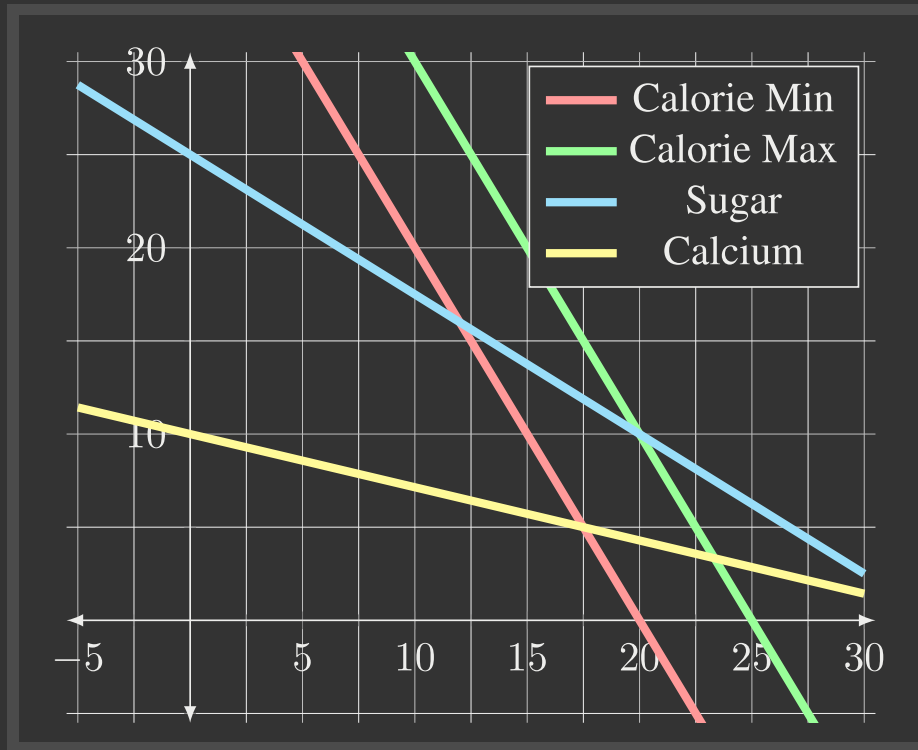
- Start with LP-relaxed version of IP
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 - Consider two branches with two different slightly more constrained LP problems:
 - Left branch: Add constraint $\mathbf{x}_i \leq \text{floor}(\mathbf{x}_{LP}^*)$
 - Right branch: Add constraint $\mathbf{x}_i \geq \text{ceil}(\mathbf{x}_{LP}^*)$
- Recursion. Stop going deeper:
 - When the LP returns a worse objective than the best feasible IP objective you have seen before.
 - When you hit an integer result from the LP
 - When LP is infeasible



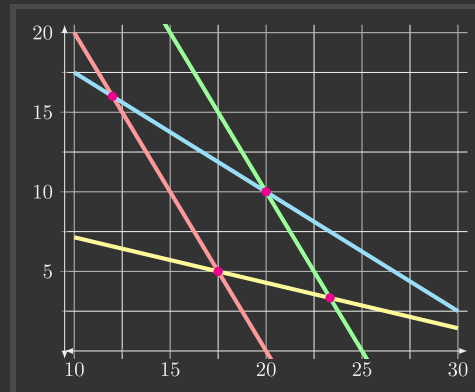
BRANCH AND BOUND EXAMPLE



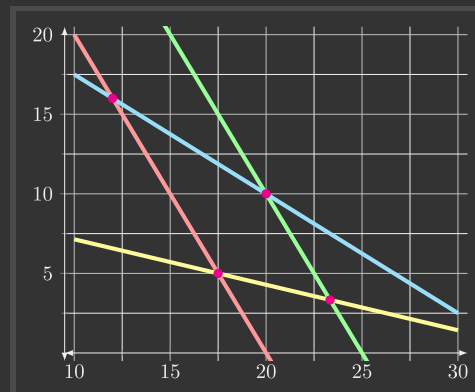
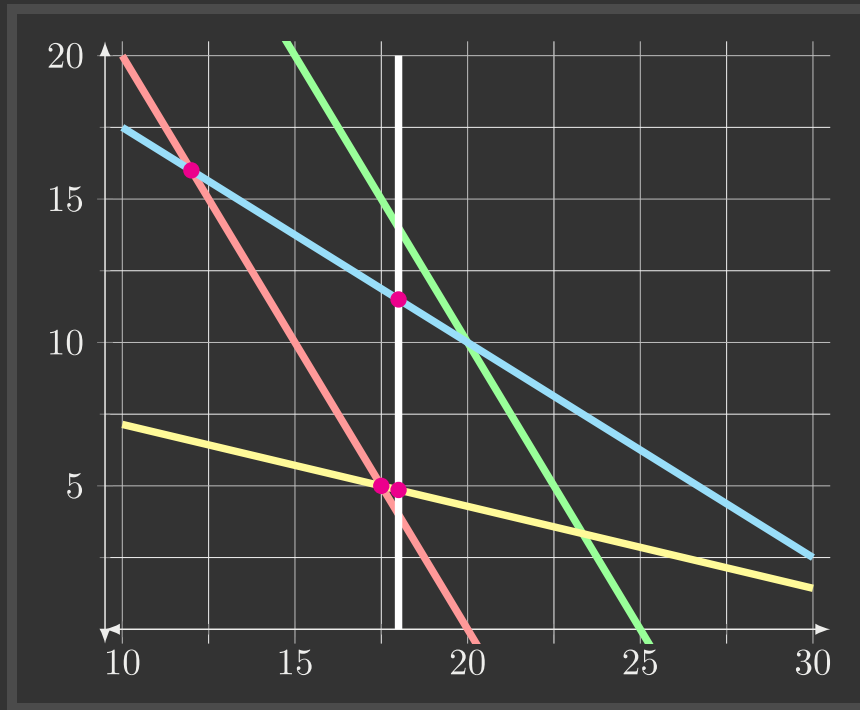
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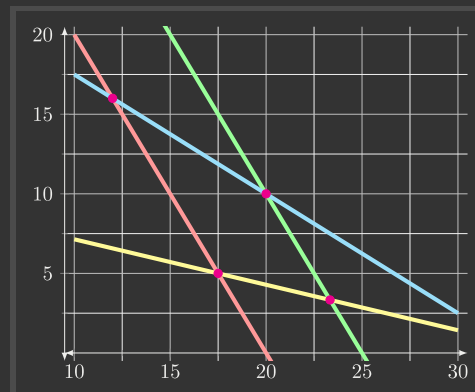
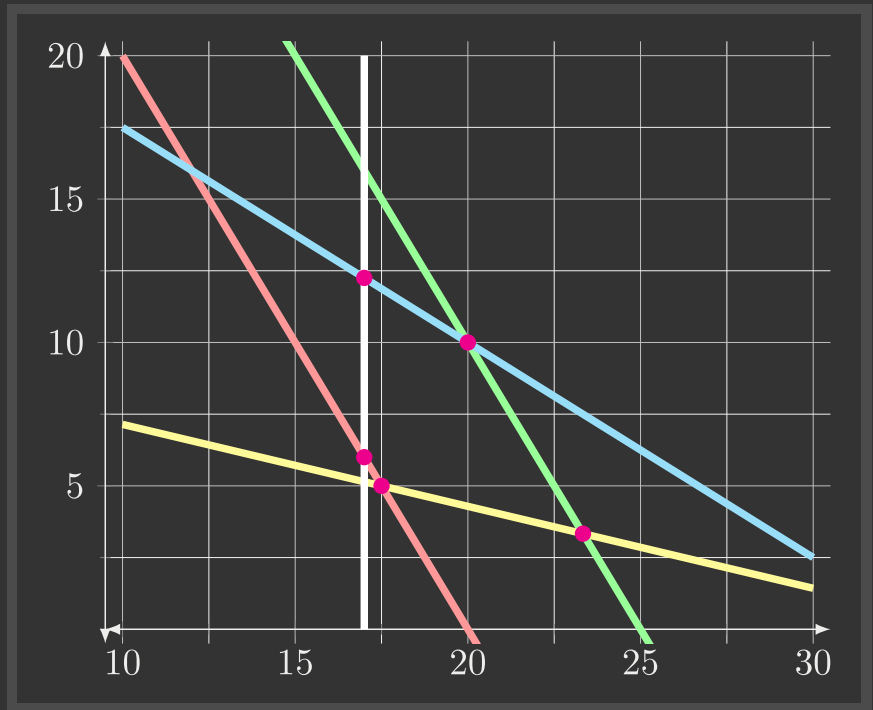
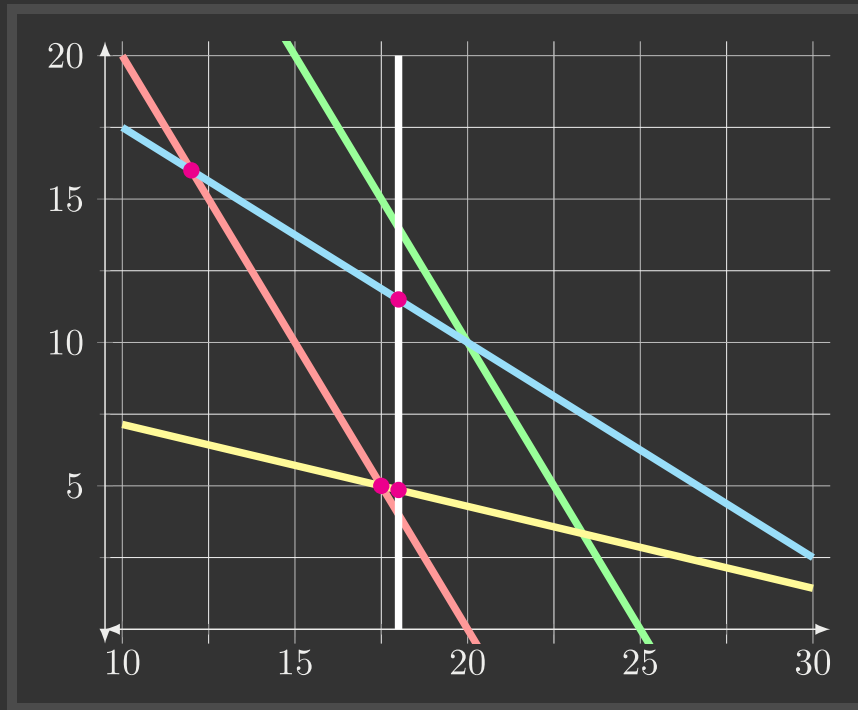
BRANCH AND BOUND EXAMPLE



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BRANCH AND BOUND EXAMPLE



Q & A



XKCD

